

VETO: Supplementary Protocol and Evidence Index

Anonymous CVPR submission

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1. Status

This is an executable-study scaffold, not a completed experimental supplement. Source code, data identities, raw answers, candidate failures, native training receipts and GPU/API costs must be linked before a result is populated.

2. Shared Scoring

Answer extraction is host-owned. Invalid ground truth is a data error; an incorrect or malformed model answer is an unsuccessful trajectory. Pair scores require complete identity-matched predictions on both sides. No invalid or missing sample is silently replaced with a zero or removed from the denominator.

3. Reproducibility

All seeds and condition names are preregistered. The same source family cannot cross data roles. Every real run retains its configuration, source snapshot, model and processor identities, decoder settings, raw responses, and terminal Slurm allocation record. Shared-prefix computations are identified explicitly. The Adam-reset restoration caveat is shared by every trained branch. The public source model is Qwen3.5-4B revision 851bf6e. All conditions start from the same audited BF16 initialization. Native RSFT stores trainable parameters in FP32 and exports BF16 serving weights. Source, derived initialization, native checkpoint and serving identities are recorded separately, with the serving export checked against the native BF16 cast.

4. Counterfactual Safety Audit

Let M_C be marginal competence and F_C the fraction of pairs with exactly one correct side. Then $P_C = M_C - F_C/2$ exactly. VETO-v3 uses jointly resampled source-paired deltas against the incoming harness. Three candidate slots, two metrics and both tails share the frozen nominal error allocation. Candidates are **PASS**, **FAIL**, or **UNCERTAIN**; the latter two are excluded and the incoming harness remains eligible. This empirical percentile-bootstrap rule has no finite-population or post-training guarantee.

The earlier C64 point gate and paired-ranking V2 are retained as historical diagnostics. V2 chose h1 over h3 by one C64 pair, but h1 was lower on the independent R512 diagnostic and its paired interval included zero. R512 is thus method-development information and cannot validate V3. Sequential repeated looks are excluded until a valid repeated-look error allocation is frozen.

5. Configuration Calibration

Prompt	Single image	Both images
Direct	83.40	78.91
Structured	88.28	83.59
Brief reasoning	86.33	81.25

Table 1. Complete initial-prompt calibration on 256 development pairs. Scores are percentages; these are not VETO gains or test results.

The structured prompt was selected by ordinary development accuracy. Learning-rate calibration is reported separately below. All formal conditions use the selected rate.

Learning rate	Single image	Both images
10^{-7}	86.52	82.81
10^{-6} (selected)	87.50	80.86
10^{-5}	54.69	34.77

Table 2. Complete learning-rate calibration on the same 256 development pairs after four native RSFT batches. Scores are percentages. These are configuration measurements, not VETO gains or independent test results.

Ordinary development accuracy determines selection; ties use the smaller rate. Seed42 shares the selected first-stage checkpoint; this calibration is not an additional independent trial.

6. Optimization Archive

Selections: WHALE h0, VETO h0, Point-estimate paired gate h0. All three rules select the same harness; this archive

Candidate	H correct	C single	C both
h0	108/128	439/512	203/256
h1	106/128	414/512	186/256
h2	63/128	240/512	93/256
h3	107/128	442/512	205/256

Table 3. Complete fixed-weight optimization archive. These are H/C selection measurements, not independent method results.

- External fixed-subset results, including adverse outcomes. 093
- Source-cluster paired intervals and per-seed raw scores. 094
- Stage traces, source-grounded success/failure cases and cost curves. 095

supplies no evidence of an independent VETO selection benefit. Unparseable answers on H/C: h0 3/8, h1 12/45, h2 55/224, h3 5/5. These descriptive counts do not identify causal visual improvements. 096

7. Completed Single-Cycle Development Trajectory 097

A complete native continuation and development reconstruction is pending. For the first seed42 proposal, the recorded tool trace read the incoming harness and attempted an absent iteration report, without reading the supplied H feedback. The three candidates are therefore generic strategy mutations, not demonstrated feedback-conditioned improvements. They were retained and evaluated without replacement. This limitation applies to the current archive. The prospective V3 search received observable H-pair outcomes and used three fixed mechanism slots: visual recheck, reasoning decomposition and response control. At the fixed seed42 post-stage-one weights, h0–h3 score 108,106,63,107 of 128 on H; 203,186,93,205 of 256 on paired C; and 439,414,240,442 of 512 marginally on C. The audit classifies h1 and h2 as **FAIL** and h3 as **UNCERTAIN**. For h3, the multiplicity-adjusted competence-difference interval is $[-0.39, 1.76]$ percentage points and the fragility-difference interval is $[-2.73, 1.95]$ points, so its favorable point estimates do not pass the frozen audit. WHALE, VETO, and the matched point-estimate gate all select h0. The preregistered distinct-selection condition therefore fails, and no V3 post-RSFT continuation is run. This complete optimization archive supplies a negative diagnostic rather than an independent method benefit.

8. Illustrative Failure and Interpretation

A source-grounded illustrative case is pending; no synthetic trajectory is substituted.

9. Required Evidence

- Main results: 18 logical conditions or explicit incomplete/failed records.
- Paired gate versus same-data marginal gate and augmentation.